

Frederick Lewin

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EDUCATION

Durham University October 2022-June 2026 (expected)
MMath in Mathematics (Third Year: 76%, Second Year: 82%, First Year: 79%)

- Fourth year modules will include stochastic analysis, uncertainty quantification, spatio-temporal statistics, and general relativity. Dissertation is on Sequential Random Absorption.
- Third year modules included stochastic processes, machine learning and neural networks, mathematical finance, Bayesian computation and modelling, mathematical biology, decision theory.

Saffron Walden County High School September 2020-July 2022

- 4 A-Levels in Further Maths (A*), Maths (A*), Physics (A) and Biology (A).

WORK EXPERIENCE

Quantitative Analyst Intern, Bank of America: (Python, C++, LaTeX) June 2025-August 2025

- Developed a low-latency calibration framework for the shifted-SABR model for pricing and risk, on the non-linear rates desk using non-linear least squares and neural networks for parameter tuning.
- Implemented unit tests within the production codebase to ensure smooth adoption of my calibration procedure, used by rates traders for hedging and time-zero pricing.
- Built an exotic TRS equity swap pricer using a copula-based correlation model, enabling efficient dynamic hedging with futures for the swaps desk.

Data Analyst Intern, Aramco Trading Limited: (Python, SQL, TensorFlow) July 2024-September 2024

- Fully-cycle ML-based ensemble model development to generate trading signals on commodity time spreads, using data from Refinitiv Eikon API and TensorFlow for model development.
- Developing the quantitative decision structure for traders and analysts by creating a live dashboard with forecasts and current data, including an automatically summarised report on weekly EIA release.
- Presenting and explaining complex pricing and forecasting models to management to encourage a growth to the quantitative decision structure in discretionary workflows.

Intern, Quant Insight: (Python, bash) August 2023-September 2023

- Investigated the average returns of trades on all US stocks over the last 5 years, found using the backtest framework, and how the returns were affected by different macro-economic drivers.
- Time-series analysis to classify trades based on the macro-economic factors with the highest change in contribution to the model value in the lead up to trading signals..

EXTRACURRICULAR ACTIVITIES

Fully-systematic HFT Strategy:

- Developed and now maintain/research fully systematic high frequency equity trading strategy, running live. Trades off of macro warranted fair value, Russell 3000 universe.
- Dynamically delta-gamma hedges the portfolio using US ETF put options to reduce beta and market exposure. PCA to reduce dimensions of factor based macroeconomic fair value models.
- Out-of-sample returns of 50.1%, volatility of 25.3% (annualised sharpe ratio of 1.98, $r=0$) hit rate of 72% since 2023-11-27. Concurrently adjusted leverage, based on market and instrument volatility.

Collingwood College Boat Club Vice President and Kit Secretary March 2024 - Present

- Oversaw the daily operations of a boat club, focusing on increasing alumni funding and engagement.
- Coordinated with external sports kit companies to design, produce and distribute kit for a club of 150+ members, incorporating suggestions to ensure the kit chosen aligns with the members needs.

SKILLS, INTERESTS AND ACHIEVEMENTS

Skills: Python, R, rstan, SQL, Bash, machine learning (sklearn, TensorFlow), algo trading, data cleaning

Interests: High frequency trading strategies, poker, chess, rowing (1V Collingwood College, Durham), competitive powerlifting, football